

TEAM HYPERION

Brisbane Boys' College - Open Soccer League - Australian National Competition



Our Team (Left to Right)

Thomas McCabe - Sensors/Software
Luke Atherton - Structural/Hardware
Sam Garg - Strategy/Software
Matthew Adams - Electrical/Hardware

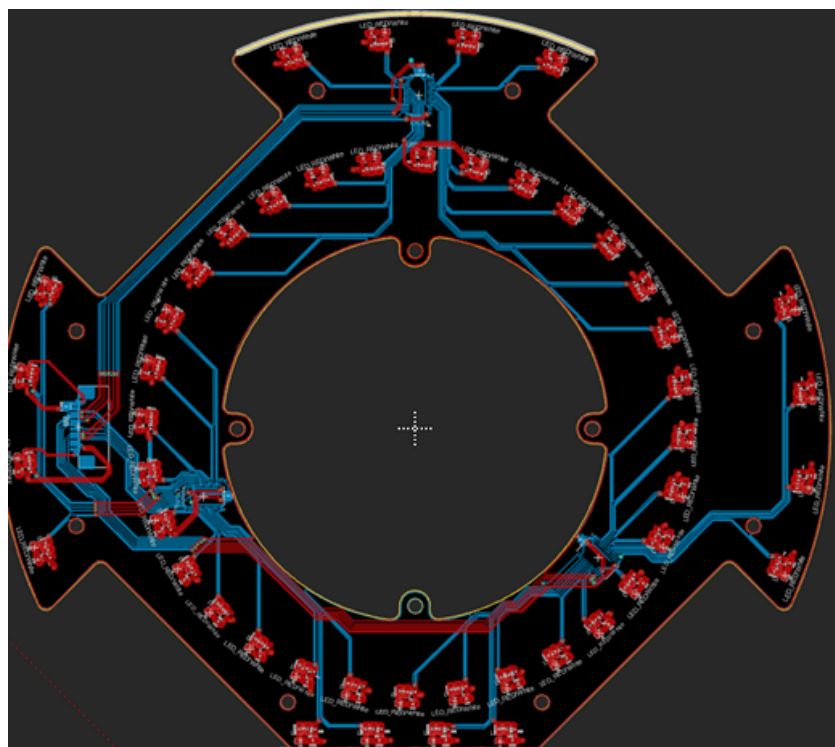
Who are we?

We are Team Hyperion, a four-student robotics team from Brisbane Boys' College, and right now, our primary focus is gearing up for the National Competition in the Open Soccer Division. We have just returned from representing Australia at the 2026 RoboCup in Incheon, South Korea, where we secured 12th Place Individually and 4th Place in SuperTeam, which proudly builds on our 2025 run in Salvador, Brazil (14th Individually, 1st Place SuperTeam). Coming off those international campaigns in the Infrared league, we are currently executing rapid technical modifications to our robot system adjusting our 360° vision processing and movement algorithms to handle the new field conditions of the Open league. After a quick trial run during the Queensland state competition, we are incredibly excited to compete in the national competition this year.

Previous Competition Experience

- 2026 Queensland States
- **2026 Internationals - South Korea**
- 2026 Brisbane Regionals
- 2025 Australian Nationals - Canberra
- 2025 Queensland States
- **2025 Internationals - Brazil**
- 2024 Australian Nationals - Queensland
- 2024 Queensland States
- 2024 Brisbane Regionals
- + Numerous Individual RoboCup Competition Experience

Our Innovations - Light Sensor Board & Kicker “Video Game” Software

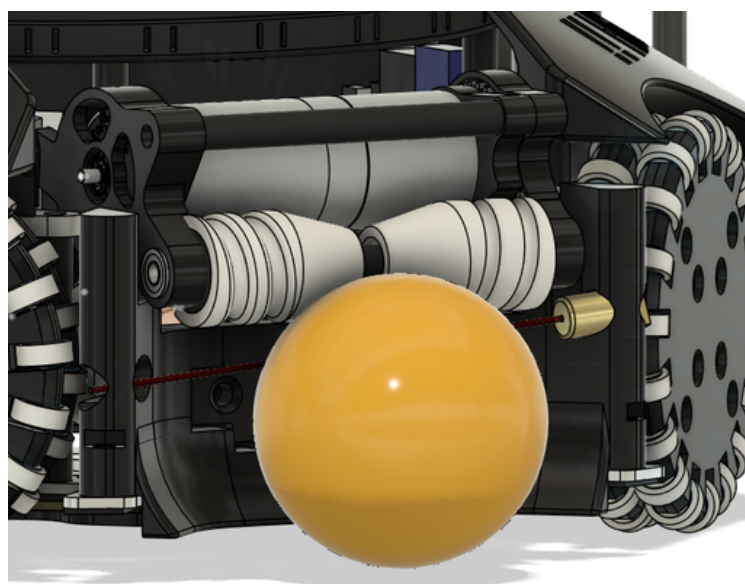


Hardware Innovation: The Light Sensor Board

The Light Sensor Board features 48 distinct component groups, each equipped with phototransistors, LEDs, resistors, and smoothing capacitors. These sensors are strategically arranged in a circular pattern ahead of the wheels to optimise a defender's positioning near the goal box, while 16 outer groups are divided into four segments between the wheels to ensure rapid line detection and response. To manage this dense array efficiently, all sensor signals are routed through three 16-bit multiplexers. This hardware configuration successfully consolidates the multiplexed data into just four analog pins, significantly reducing the number of processor pins required for operation.

Software Innovations: The Ball Handling “Video Game” Software

The Kicker Software operates almost like a video game. This coded library coordinates the robot's ability to safely shoot the ball automatically, acting as a strict “safety checklist” to prevent weak, accidental, or dangerous solenoid triggers. For example, one checklist item ensures the kicker has over 40V of stored power before firing. Just like an energy meter in a game, the robot holds a maximum of 5 kicks at any given time. When it does kick, it requires 20 seconds to recharge before that kick can be replenished, which prevents the hardware from overheating and causing a potential safety hazard. Additionally, the software enforces a mandatory 5s cooldown timer after every kick to allow the electronics to safely recover. To precisely time the shot, the kicker system uses a light gate made from a simple laser pointed directly into a light-dependent resistor (LDR). When a ball rolls into the capture zone, it blocks the beam of light from reaching the sensor. This causes the LDR's reading to drop, instantly signaling to the system that the ball has entered the capture zone. Finally, to ensure maximum hardware safety, a small LED on top of the robot's PCB illuminates whenever there is voltage stored within the capacitors. If the capacitors still hold a charge when the robot is turned off, the system's fuse may blow. This warning LED lets team members know exactly when voltage is still present, ensuring the capacitors can be safely drained before powering down the robot.

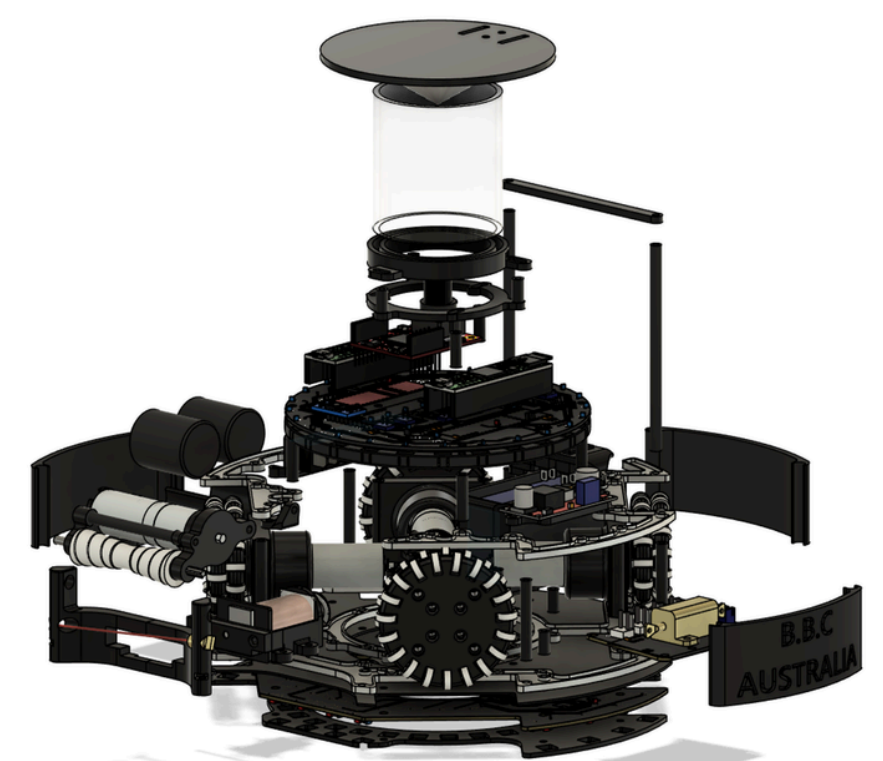


Our Challenge – Ball Control & Field Vision

The Challenge: The robot struggled to maintain ball control and create scoring opportunities. Our previous momentum-based approach was ineffective under the international field restrictions and caused gradual damage through collisions with opponents. Additionally, the reflective-foil mirror used for goal and ball tracking was unreliable. Its steep viewing angle and surface warping created significant blind spots, limiting the camera's ability to consistently detect and track the ball and goals.

The Solution: The team engineered a custom kicker and dribbler system. The dribbler uses a 3D-printed housing with a 1:2.81 speed-increasing gear ratio and high-grip silicone roller for reliable ball control. The kicker uses a modified TAKAHA solenoid with a custom hand-wound coil, lightweight 3D-printed mount, and wide plunger tip for consistent contact across the capture zone. A voltage booster increases the supply from 12 V to ~48 V, substantially increasing kicking power, while an integrated photogate detects ball capture.

A custom CNC-machined hyperbolic aluminium mirror was manufactured and polished. Its reflective curvature was calculated in Desmos based on the robot's dimensions and desired viewing range.



What have we learnt throughout our time at RoboCup?

Throughout our time at RoboCup, we have experienced highs and lows. Whether that was the unparalleled thrill of placing in a competition, or the absolute panic of our robot's wheels disintegrating right before a match, we have experienced a lot of what RoboCup has to offer. As a result of this, there are many things that we have learnt. This includes valuable technical and coding skills that we can take into our future professions and use to stand out in job applications. Moving beyond basic programming, we evolved into ‘multidisciplinary’ engineers. Structurally, we learned how to design resilient robots and account for the brutal physical forces our robots would face on the field. Electrically, we tackled power distribution, soldered reliable connections, and ensured clean signal integrity between components. By combining this hardware knowledge with complex software logic to process sensor data in milliseconds, we truly learned how to bridge the gap between digital code and the unpredictable physical world. However, the most profound lessons haven't just been technical. We have also grown significantly as a team over the years, which has given us the valuable ability to communicate, collaborate, and thrive under immense pressure. We learned how to divide tasks based on our individual strengths, trust one another's expertise, and navigate the high-stress environment of competition days by turning disagreements over design choices into constructive compromises. Most importantly, watching our hard work occasionally fall apart on the mat taught us true resilience. We learned to treat failures not as personal defeats, but as crucial data points to improve our iterative design. Ultimately, RoboCup was never just about making a robot function; it was a pressure cooker that forged us into better problem solvers and a more cohesive team, building a foundation of grit and synergy that will outlast any tournament.



Our Website